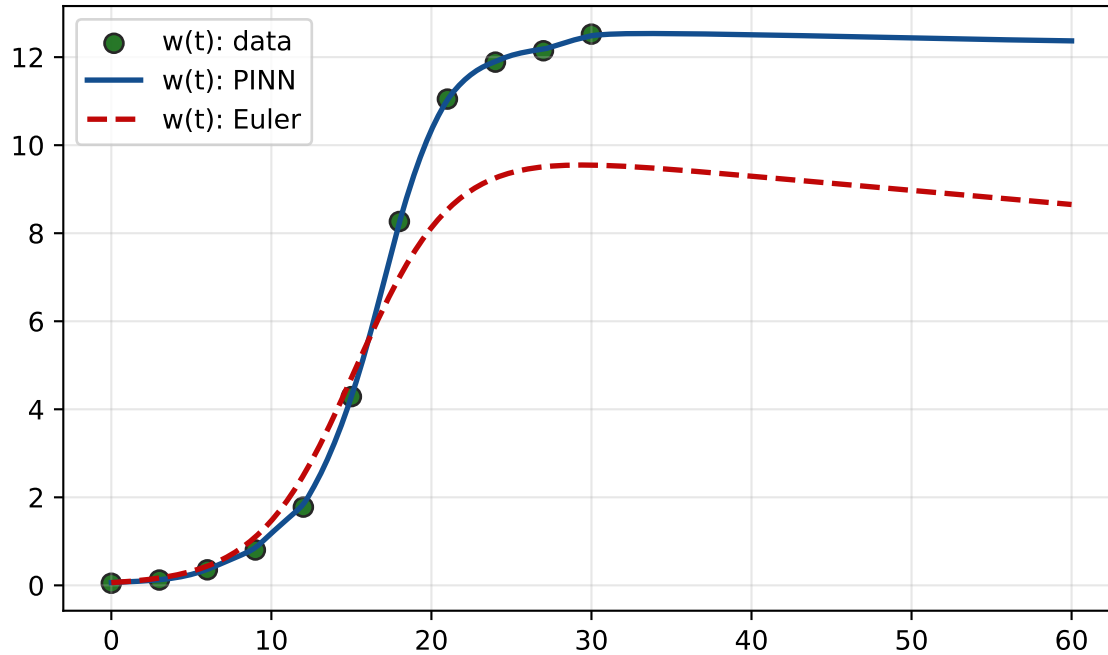
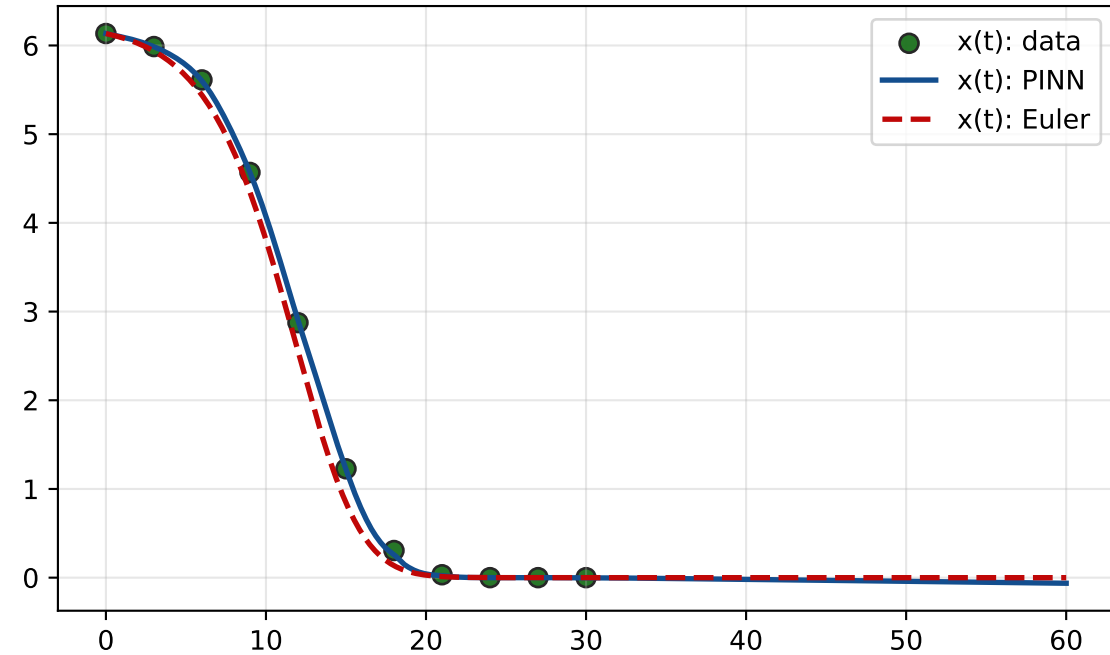


# E10 | PINN vs Euler

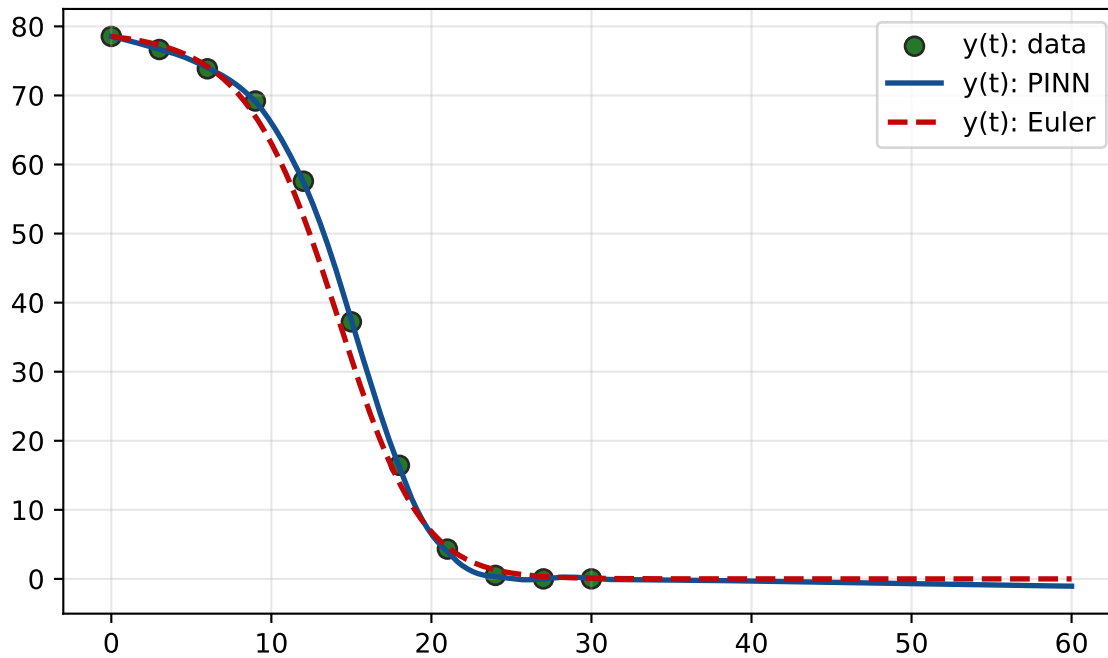
w(t): Biomass



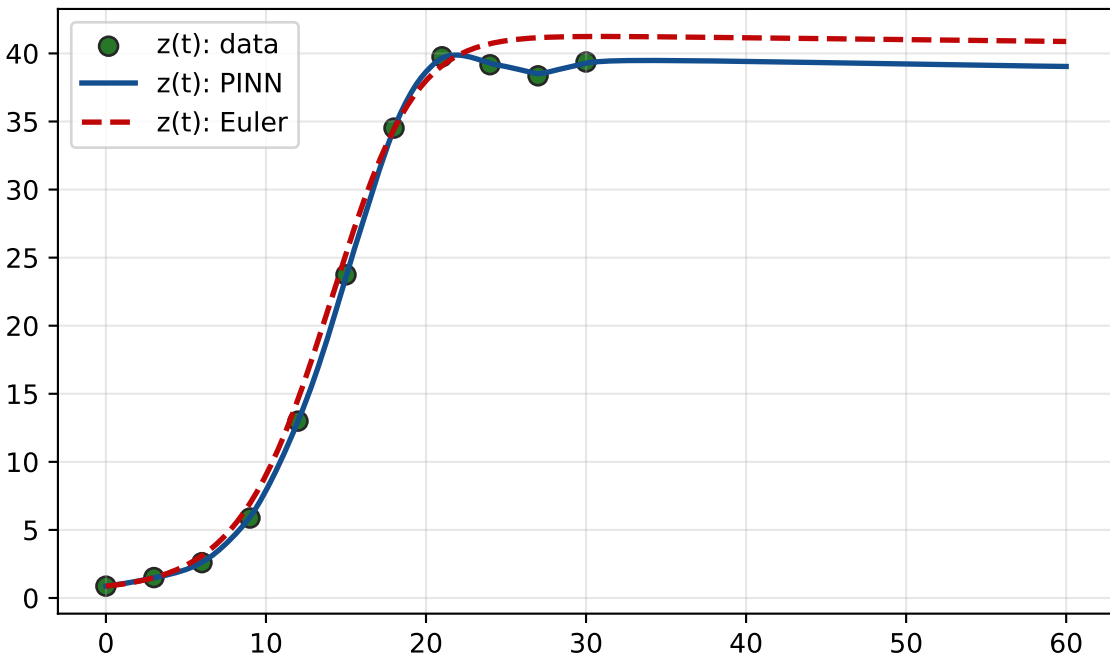
x(t): Glucose



y(t): Fructose



z(t): Ethanol



E10

R2=0.9920, R2\_adj=0.9907, RSS=1.202e+02, AIC=5.851e+01

cond(JTJ)=4.92e+05, used\_pinv=False

| param | estimate     | SE           | MOE_95       | CI_low        | CI_high      | p_value      |
|-------|--------------|--------------|--------------|---------------|--------------|--------------|
| p1    | 3.237425e-01 | 2.037023e-01 | 4.123737e-01 | -8.863120e-02 | 7.361162e-01 | 1.202810e-01 |
| p2    | 3.245131e-02 | 1.516348e-02 | 3.069686e-02 | 1.754445e-03  | 6.314817e-02 | 3.882811e-02 |
| p3    | 1.000000e-02 | 2.174470e-01 | 4.401985e-01 | -4.301985e-01 | 4.501985e-01 | 9.635605e-01 |
| p5    | 1.035071e-01 | 5.908409e-02 | 1.196095e-01 | -1.610244e-02 | 2.231166e-01 | 8.786555e-02 |
| p7    | 4.681240e-02 | 8.552037e-03 | 1.731269e-02 | 2.949970e-02  | 6.412509e-02 | 2.998512e-06 |
| p9    | 2.353127e-02 | 4.250419e-03 | 8.604524e-03 | 1.492674e-02  | 3.213579e-02 | 2.462931e-06 |

| IC             | estimate     |
|----------------|--------------|
| w0 (estimated) | 6.204622e-02 |
| x0 (fixed)     | 6.134528e+00 |
| y0 (fixed)     | 7.854383e+01 |
| z0 (fixed)     | 8.686359e-01 |
| W0 (fixed)     | 0.000000e+00 |